PHY · Cosmology — syllabus points and past-paper evidence

Every award of the 2016–2025 papers, labelled to the 2026/2027 Year 12 syllabus content (identical in both). Statistics from the labels; notes written by AI from that evidence: review before use.

Pattern 1: Astronomical distances: AU, light year, parsec

Syllabus: S26-U4-SU-35 / S27-U4-SU-35

Frequency: 3 of 10 exam years; 4 marks in all.

Marks reward: Final answer 2, Select the principle or relationship 1, Substitution and working 1

Past questions: 2016C-21(a)； 2020P-19(ci)； 2023R-8

Summary: This point tests whether you can express large astronomical distances in astronomical units (AU), light years (ly) and parsecs (pc), and convert between them and into kilometres. It usually appears as short conversion or light-travel-time questions, sometimes linked to a Hubble-law graph of recessional speed against distance.

Recognise: Look for a distance given in ly, pc, Mpc or AU that must be converted, a question asking how long light takes to arrive (answer in Earth years), or a Hubble-type graph where the distance must be read off and then changed into kilometres. Any question that says 'convert' or asks for a value in a different astronomical unit is this point.

Steps

1. Write down the given value with its unit and the unit the question asks for, keeping powers of ten explicit.
2. Take the conversion factors from the data sheet rather than memory: 1 pc is about 3.26 ly, 1 Mpc is 10^6 pc, 1 AU is about 1.50 × 10^8 km, and 1 ly is the distance light travels in one year (about 9.46 × 10^12 km).
3. Convert by multiplying or dividing by the factor in the right direction: a larger unit gives a smaller number, so ly to pc divides by 3.26 and pc to ly multiplies by it.
4. For light-travel time, note that light covers 1 ly in 1 year, so the travel time in Earth years has the same numerical value as the distance in light years.
5. For a Hubble-law distance, find the recessional speed as a fraction of c (c = 3.00 × 10^5 km s^-1), read or calculate the distance in Mpc from the graph (d = v / H0), then convert Mpc to km using 1 pc ≈ 3.09 × 10^13 km.
6. Check the answer: a distance in km should be very large (many orders of magnitude bigger than AU values), and the unit must be written with the number.

Where the marks are

- A mark for the final answer with the correct unit, for example years, Mpc or km.
- A mark for stating or applying the correct conversion factor from the data sheet, shown before the calculation.
- A mark for accurate arithmetic with powers of ten when chaining a conversion, such as Mpc to pc to km.
- A mark for correctly using the light-year idea, for example that light travel time in years equals distance in ly.

Pitfalls

- Confusing light years with parsecs, or using 1 pc = 1 ly, which is a factor of about 3.26 out.
- Forgetting the 10^6 prefix when converting between Mpc and pc, or mis-keying the exponent on a calculator.
- Giving the answer as a time in seconds or years without checking that it matches the question's unit.
- Multiplying in the wrong direction when converting, so the answer is smaller or larger than the physical distance.

- Using the recessional speed in km s^-1 directly as a distance, or using c in the wrong unit when finding the fraction of light speed.

Pattern 2: The Big Bang and the early universe

Syllabus: S26-U4-SU-36 / S27-U4-SU-36

Frequency: 3 of 10 exam years; 4 marks in all.

Marks reward: Qualitative explanation 2, Final answer 1, Recall 1

Past questions: 2017P-16(c)； 2018C-18(d)； 2021R-12(a)

Summary: This point asks you to describe what the Big Bang theory says about the early development of the universe and the evidence that supports it, mainly Hubble's law and the cosmic microwave background (CMB). It usually appears as short explain questions, with occasional multiple-choice items on the early sequence of events.

Recognise: Look for the words Big Bang, early universe, expansion, Hubble's law, galaxies receding, redshift, singularity, cosmic microwave background, or a question about which forces existed just after the Big Bang.

Steps

1. Decide what the question wants: a claim of the theory, a piece of evidence for it, or the early sequence of events.
2. For Hubble's law, use v = H0 x d: more distant galaxies recede faster, so the universe is expanding.
3. Explain support for the Big Bang by running the expansion backwards: all galaxies trace back to one point, a singularity, which is the theory's starting state.
4. For the CMB, state that the early universe was extremely hot and dense, and that radiation was released once atoms could form.
5. Explain that as the universe expanded, that radiation cooled and stretched to microwave wavelengths, so it is seen in all directions with almost the same temperature.
6. For the early forces, link the sequence to falling temperature: at extremely high temperature the forces were unified, and they separated as the universe cooled. Use the order given in your course notes.
7. Write each point as a full sentence joined by 'because' or 'so', tying the evidence back to the theory.

Where the marks are

- A mark for stating that the early universe was very hot and high in energy.
- A mark for explaining why the evidence supports the theory, not just naming the evidence.
- A mark for linking the backward trace of galaxy motion to a single starting point.
- A mark for the CMB linking early high temperature and radiation to the uniform microwave background observed today.
- For multiple choice on forces, one mark for selecting the single option that matches the order in your notes, with no ambiguity.

Pitfalls

- Saying Hubble's law proves the Big Bang; it supports it.
- Describing the Big Bang as an explosion of matter moving into space that already existed.
- Stating that the CMB comes from one direction or from stars; it is a near-uniform background.
- Naming a term such as redshift or CMB without explaining the cooling or expansion that causes it.
- Mixing up the order of the forces, or choosing by memory of a headline rather than the sequence in the notes.

Pattern 3: Evidence for the Big Bang

Syllabus: S26-U4-SU-37 / S27-U4-SU-37

Frequency: 6 of 10 exam years; 28 marks in all.

Marks reward: Qualitative explanation 17, Final answer 3, Graph work 3, Substitution and working 3, Recall 2

Past questions: 2016R-4(c); 2016R-4(d); 2018C-18(c); 2018C-18(d); 2020P-19(b); 2020P-19(cii); 2020P-19(d); 2021R-12(a); 2021R-12(b); 2023R-8; 2024R-11(a); 2024R-11(b)

Summary: This point asks students to link the main evidence for the Big Bang (cosmic background radiation, expansion of space, light-element abundance, and red shift of distant galaxies obeying Hubble's law) and to apply v = H0 d. In past papers it mostly appears as explain or describe questions on red and blue shift and what Hubble's law shows, with occasional short calculation or graph-reading parts; light-element abundance and background radiation appear less often.

Recognise: Look for the words Big Bang, cosmic background radiation, red shift, blue shift, Hubble's law, recessional velocity, or galaxies moving apart. Questions often give a spectrum, a velocity-distance graph, or a Doppler equation and ask what the observation shows about expansion, or ask you to find a velocity or distance from v = H0 d.

Steps

1. State the observation precisely: spectral lines from distant galaxies are shifted towards longer wavelengths (red shift), or the background radiation is observed at longer wavelengths and lower energy than when it was emitted.
2. Explain the cause: as space itself expands, the wavelength of light travelling through it stretches, so a receding source gives red shift and an approaching source gives blue shift.
3. Link to distance: the further a galaxy is, the larger its red shift and the faster it recedes, so recessional velocity is proportional to distance.
4. Apply Hubble's law v = H0 d, rearranging to H0 = v/d or d = v/H0 as needed, keeping units consistent (for example km s-1 and Mpc).
5. For a Doppler calculation, use delta-lambda / lambda0 = v/c to find the wavelength shift, then add it to the emitted wavelength to get the observed wavelength.
6. Conclude: galaxies are moving apart in all directions, so the universe is expanding; tracing this back implies all matter came from a single origin, which supports the Big Bang. The background radiation is the cooled, stretched relic of early radiation.

Where the marks are

- Stating the observed feature exactly (lines shifted to the red end, or greater shift for greater distance) earns the knowledge mark.
- A mark for explaining red shift as wavelength stretching or recession, linked to space itself expanding.
- A mark for stating the proportionality: further galaxies show greater red shift and recede faster.
- Calculation and graph questions: a mark for the correct formula with consistent units, a mark for the correct intermediate value, and a mark for the final answer with units (carried forward from earlier errors); for graphs, a mark for locating the value on the correct axis, reading the value, or a gradient from two chosen points on the line of best fit.
- A final mark is usually given for a concluding statement that ties the evidence to uniform expansion or to a single Big Bang origin.

Pitfalls

- Swapping red and blue shift, or saying an approaching galaxy is red-shifted.
- Describing a wavelength change without saying whether it increases or decreases.
- Saying galaxies move through space, rather than space itself expanding, which is the basis of Hubble's law's interpretation.
- Unit errors: mixing km s-1 with m s-1, Mpc with km, nm with m, or failing to convert a percentage of c into a speed.
- Stopping at the observation without concluding about expansion or the Big Bang, or using Hubble's law without linking it to a single origin.